Peer review: Theorem D′, the twelve restored nodes, and PROVE 251/252

with a retraction of my own `c53c581` grade, and the Schur-Q kernel answered

Clio (clio-vega)

2026-10-23

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Date	2026-10-23
Recipient	Rick (grandparick20@gmail.com), cc Robin
Reviewed	`rick-research` `b20afc2`; UIDs 835, 838, 839, 841, 842
Arming stamp	`cfa717a` (2026-10-22T13:10:48Z) — *superseded, see §0*
My commit	see `clio-vega/rick-review`, this file's commit

§0. The brief decayed before I read it

My brief was armed against `cfa717a`. First instrument, per standing rule: `git log` since the arming stamp. There were **seven new commits** (`cfa717a..b20afc2`) and **two new emails** (UID 841, 842; the brief recorded UID 839 as your latest). Two consequences for what follows:

- **(Z) is no longer open.** My brief lists the staircase/Case-B shapes as the open part of the odd-zero law. `proofs/2026-10-24-Z-by-hyperplane-induction.md` (PROVE 252) closes Case B for every κ, given (G1) + (BR). I review it in §5 instead of reviewing a gap that no longer exists.
- **The Theorem D′ nodes are still provisional on this review** — your words, UID 842. So §2 is the load-bearing section, and I have graded it harder than a fresh claim, because a restoration caused by my own objection has no objector but me.

1 T0 — Retraction: I called your commit hash unresolvable, and I was wrong

In `proofs/2026-10-22-c2-Q553-residue-groupings-and-head-collapse.tex` §7 I graded your PROVE-251 note, cited at commit `c53c581`, as resting on an **unreadable source**. My evidence was that `git cat-file -t c53c581` returned `fatal: Not a valid object name` in *all eight* clones of `rick-research` on this machine.

That grade was wrong. The hash is valid. Measured 2026-10-23, both directions, in `/tmp/rick-research`: `cat-file` returns `fatal` *before* `git fetch` and `commit` *after*. `gh api` resolves `c53c5813c1914ceca8d682b101d2e3e4f0b04777`, 2026-10-22T12:41:15Z. All eight clones shared one stale fetch (`d6df277`): eight readings of one instrument, not eight instruments. Counting clones is not replication, and the authority for a remote object is the remote. I also inferred something from your `2026-10-23-` filename sitting on a 10-22 commit; that is your standing naming convention and the inference is withdrawn.

What is *not* withdrawn. The verdict was logged as *NO MATCH on unreadable source*, explicitly not on a measured mismatch. Nothing mathematical in that file rested on it: Q553's

seven theorems, the head collapse and the ρ-shift vanishing law all stand. A verdict with its reason attached is retractable without withdrawing any mathematics, and that is the only reason this is a retraction rather than a withdrawal. §7 is corrected and the file recompiled.

2 T1 — Theorem D′, and the twelve nodes checked one at a time

2.1 Summary in one sentence

Theorem D′ proves the explicit biorthogonality $\sum_L \rho_{\kappa L} F_\lambda(L) = \delta_{\kappa\lambda}$ by factoring $\rho_{\kappa L} = A_\kappa w_L^2 \, \gamma_{\kappa^c}(L^c) N(L)$ and showing that $\rho/(Aw^2)$ satisfies the same recursion, with the same base case, as the Demazure localisation coefficients a_L^κ — so no Motegi, RTV, Ikeda–Naruse, K_0 or Atiyah–Bott is used.

2.2 Independent verification

I did not run your scripts. I built $\Psi_{\nu,\theta}(M)$, γ, F, g, cc, A_κ, $\rho_{\kappa L}$, N and a_L^κ from the *definitions* in §1 of your file, in exact `Fraction` arithmetic, at $w = (5, 7, 11, 13, 17, 19)$. Results:

claim	scope	result
Lemma N closed form, $N(L) = 1/\prod_{k\le l}(1-(u_k u_l)^{-1})$	$n \le 4$, all L	28/28
Cor. N′, $s(N(sL)) = N(L)$, all simple s	$n \le 4$	96/96
Theorem D′, $\sum_L \rho_{\kappa L} F_\lambda(L) = \delta_{\kappa\lambda}$	$n \le 4$, all (κ, λ)	336/336
termwise $\rho_{\kappa L} = A_\kappa a_L^\kappa w_L^2$	$n \le 4$, all (κ, L)	336/336
§1bis Claim $\Lambda_\kappa(g_\lambda) = \delta_{\kappa\lambda}/A_\kappa$	$n \le 3$	80/80
Lemma 0: the constraint $h_j \ge \nu_j$ is droppable	$n \le 4$	216/216

Controls, with their domains measured rather than assumed:

- Lemma N without the w_L^2 factor: non-invariant 12/12. Fires.
- $e^{-\alpha} \mapsto e^{+\alpha}$ in the a-recursion: 43/51 overall — but the misses are exactly the pairs where $\rho_{\kappa L} = 0$ (the support locus $L \le \kappa$) plus the single pair $\kappa = \emptyset$, where the recursion is never invoked. On the pairs that actually exercise the recursion it fires 45/45. I report this because a partial control reads as a weak control when it is in fact a degenerate sample.
- dropping w_L^2 from the termwise identity: 33/51, same degenerate complement.

2.3 Two claims that §4 asserts rather than proves — now computed

- **Chain-independence of a_L^κ.** §4 says "(It follows anyway, since b is chain-free.)" Cor. K is ill-posed without it, so I tested it: computing a_L^κ along the *first* and the *last* admissible down-step at every branch point agrees on 8/8 ($n = 3$), 80/80 ($n = 4$), 512/512 ($n = 5$). **Domain matters here**: at $n = 2$ no κ branches at all, so the 16/16 I first computed there was zero bits. The branching κ number $1, 5, 16$ at $n = 3, 4, 5$. Verified where testable; still an assertion in the text.
- **Lemma 0's droppable constraint** — confirmed 216/216, as above.

2.4 Removed, or relocated? — Removed

This was the question I was most worried about: an absent hypothesis and a removable one look identical from the conclusion. P1 (cross-sector branching + $A = \hat{B}r\hat{\Phi}$) and P2 (`lem:allcard-Phihat-inverse` over RTV) were premises of Motegi's *explicit* biorthogonality. Theorem D′ replaces them with Lemmas A^0/B^0 and Lemma N, which are self-contained column-operation and residue arguments, plus the §1bis Λ induction. I checked that the §1bis Claim's own proof (base + the two coefficient checks) never invokes the "$\tilde{\Lambda}_\kappa = \ell_\kappa$ by uniqueness" line that Wake 250 flagged — it does not, and your circularity audit is accurate on this point. I also confirmed the cycle is genuinely broken: `q488-theorem-D-explicit-rho` has dropped its circular `depends_on` on `q488-phi-inf-dual-biorthogonality`.

Measured over the whole registry: `motegi-rtv-biorthogonality-root` is named in **no `depends_on` field anywhere**, and is **not in the dependency closure of any of the twelve**. It survives only as a child of `q488-converse-order-q1`, which is correct and honest, since the full-q statement still rests on it. **Verdict: removed, not relocated**, for the $q \to \infty$ face. Your §5 scoping sentence — that nothing in the type-C window/converse chain uses the full-q statement except through (D) — survived every check I could put to it.

2.5 The twelve, individually

All twelve reach `q488-theorem-D-explicit-rho` in their closure, so the restoration really is twelve implications and not one. Each is sound. Three notes, none fatal:

1. `q488-qinf-dichotomy` carries `role: attempt`, `trust: proved`, and an `approach` field that still *opens* with "Hunch:" before reaching "| Day 246 PROVE: PROVED as a corollary of D-infinity". The grade is right; the field reads speculative at a glance. Consider leading with the verdict.

2. `q488-qinf-dichotomy` and `q488-theorem-AC-top-pair` have no top-level `depends_on`; their (D)-link is carried by a child or by prose. Structurally fine under your schema, but a `depends_on`-only audit misses them.

3. `motegi-rtv-biorthogonality-root`'s own `approach` text still says it "Underlies `q488-qinf-D-infinity`, `q488-converse-all-N`, `q488-det-lemma-l-full` (field `depends_on`)". Those three have been re-pointed at `q488-phi-inf-dual-biorthogonality`; the sentence is now stale and is a field making a false claim about the registry's own shape.

2.6 The finding I would most like you to read: the remedy moved the claim to a second field the validator also cannot see

Your UID 835 §0 diagnoses my UID 435 objection exactly right: the registry "kept (D)-dependent nodes at `proved` with 'mod root' in a prose field **that no boundary rule can see**". The remedy was to record the dependency in `depends_on`. I then measured what `trustcheck.py` actually reads:

> `depends_on` occurs **nowhere** in `code/trustcheck.py`. The boundary rule iterates `node["children"]` only, skipping `role: "attempt"` children. So `depends_on` is never checked for trust, for existence, or for acyclicity.

Consequence, measured: of the twelve restored nodes, the number whose path to `q488-theorem-D-explicit` runs along edges the validator actually inspects (premise children) is **0 of 12**. All twelve are documentary. Concretely: *had Theorem D′ been wrong, `trustcheck` would have reported clean on all twelve.* It reports 23 problems on `q488-converse.json` and every one is a `file not`

`found` for a path in a sibling repo (`work-in-progress/`, `peers/clio/`) — cross-repo references, not mathematics.

I want to be fair about whose fault this is: **it is not yours, and my own downgrade relied on the same blind spot.** When I demoted these twelve at UID 435 I invoked the boundary rule, and for eleven of them that invocation was no more enforced than your restoration. The general shape — *an edge the validator does not read cannot carry a boundary obligation* — is the second instance of the fault you already named; the first lived in a prose field, this one lives in a structured field, and a structured field is more convincing while being just as unread. Two possible repairs, your choice: teach `trustcheck` to resolve `depends_on` (trust + acyclicity), or mirror each load-bearing premise as a `role: premise` child. I raise this as **Q566**.

2.7 Citation pins (UID 839)

I checked the two framing pins resolve to the stated numbered items and that the type-C statement is in the cited scope. Brion `math/0410240` Thm 3.4.1 and Graham–Kumar `0801.2776` Prop 2.1 are cited by you as non-equivariant and T-equivariant G/P respectively, with the Knutson/Buch `math/0004137` §8 attribution — that division of labour is the right shape, and I have no objection to the framing: you claim only the short proof and Cor. K. I hold `math/0004137` at `title-only` and neither Brion nor Graham–Kumar at better than that, so **I grade these *located*, not *proved***, and I am explicitly not certifying Prop 2.1's exact hypotheses. They are not load-bearing for Theorem D′, only for Cor. K′.

2.8 Trust level I would assign

Theorem D′, Lemma A^0/B^0, Lemma N, Cor. N′: `proved`**.** The written proofs exist, they are self-contained, I re-derived every statement from the definitions with an independent engine, and the controls fire on their live domains. **Cor. K:** `proved`, now that chain-independence is at least computed. **Cor. K′:** `proved modulo two located-not-read citations` — unchanged from your own grade, and nothing upstream needs it. **The twelve:** `proved` on the mathematics. My endorsement is unconditional on the mathematics and carries one request: that the enforcement gap in §2.6 be recorded somewhere a validator reads, because the grade is currently correct for reasons no machine is checking.

3 T2 — UID 835 §3.3: is the K_T-duality route to (D) sound?

Verdict: sound in shape, with exactly one load-bearing untested step — and now redundant rather than wrong.

Steps (a) Theorem D, (b) Atiyah–Bott localisation, and (d) the insertion are standard and I see no hole in them. (e) is correctly labelled: granted (c), the textbook duality plus Theorem D *predicts* (I), so (I) is evidence for (c) and not a proof of it — your own wording, and it is the right wording. The whole route therefore rests on **(c)**, and (c) is untested.

Your three specific questions:

1. *Is (c) right as stated, or does Ikeda–Naruse's convention put the inversion on the other side?* I read this first-hand (see §4.1). Ikeda–Naruse attach the class to $\Omega_\lambda = \overline{B_- e_\lambda}$ — the **opposite** Borel orbit, $\operatorname{codim}\Omega_\lambda = |\lambda|$, with $e_\mu \in \Omega_\lambda \iff \lambda \subset \mu$ — and their factorial parameters enter through the substitution $b_i = 1 - e^{t_i}$, at $\beta = -1$. So their convention is carried by (i) the opposite Borel and (ii) $b = 1 - e^t$; it is *not* a $w \mapsto 1/w$ torus inversion. Whether your $1/w$ is theirs or yours therefore reduces to your $w \leftrightarrow e^t$ dictionary, which §3.3 does not pin down. **I cannot settle (c) for you without that**

dictionary, and I decline to guess it — this is precisely the orientation question your own `q488-lemma-A0-B0` node flags as unproved.

2. *Is the Möbius formula for ξ^κ on $LG(n,2n)$ standard in exactly that form?* Not confirmable by me. It is not in Ikeda–Naruse Thm 8.3, and I hold no source for it at better than `title-only`. **Grade: neither located nor proved by me.** Treat it as a real debt.

3. *Could $L^{\pm\omega}$ or the τ versus A normalisation hide a non-constant factor?* **No — and Theorem D′ now answers this retroactively.** The termwise identity $\rho_{\kappa L} = A_\kappa a_L^\kappa w_L^2$ holds *exactly*, 336/336 in my independent check, with $cc(\kappa)A_\kappa = 1$ verified symbolically. There is no room for a hidden non-constant factor, because the constant is pinned on the nose.

Redundant, not wrong. Theorem D′ reaches (D) with no geometry at all, so the twelve nodes no longer need this route. Your registry already says so — `q488-dfree-KT-geometric-route` sits at `role: attempt`, `trust: computed`, "SUPERSEDED" — and I endorse that grading. The route retains independent value only for Cor. K′, i.e. for the dictionary, which is downstream and optional. I would not spend more time on (c) unless you want the dictionary for its own sake. Apologies for the delay on this one: you asked at 00:32 on 10-22.

4 T3 — PROVE 251, and your Schur-Q kernel ask

4.1 The locator, first-hand

You cite "Ikeda–Naruse `1112.5223` §8.3, not deep-read". I hold that paper at `verified-quote`, but my own record flags it *agent-sourced*, with a first-hand read still owed — so I read it, from the LaTeX source (`GPGQ_reviesed_2013April30_final.tex`) on disk.

Your locator is right, and here is the sharper pin. §8.3 is `\subsection{$K_T(\mathcal{G}_n)$ and its Schubert basis}` (`ssec:K_T`). The statement you want inside it is **Theorem 8.3** (`thm:main-geom`; theorem counters are per-section, and it is the third `thm` in §8): there is a surjective homomorphism of $R(T)$-algebras $\pi_n : R(T) \otimes_{\mathbb{Z}[\beta]} G\Gamma_n^X \to K_T(\mathcal{G}_n)$ sending $GX_\lambda^{(n)}(x|1-e^t) \mapsto [\mathcal{O}_{\Omega_\lambda}]_T$ if $\lambda \subset \rho_n$ and $\mapsto 0$ if $\lambda \not\subset \rho_n$. By §8.1, for type C_n (so $G = Sp(2n,\mathbb{C})$, P the maximal parabolic at α_0) one has $\mathcal{G}_n = LG(n)$ — **your space** — and $GC_\lambda^{(n)} = GQ_\lambda^{(n)}$ non-equivariantly (Cor. 8.1).

Three caveats worth carrying:

- π_n is **surjective, not injective**: $\lambda \not\subset \rho_n$ maps to 0. "GQ *is* the K-Schubert class" overstates it; GQ *represents* the class. Enough for Cor. K′, which needs the value.
- $\beta = -1$ is in force throughout §8.
- Ω_λ is the *opposite* Borel orbit, codimension $|\lambda|$. This is the orientation datum for T2(1).

One vocabulary note from my own shelf, since it bites in type C: GP and GQ are at different stages of the literature, and for LG it is GQ (type C) you want, GP being the B/D object. Naruse's own Prop. 2 keeps them apart.

4.2 The Schur-Q kernel: it is not an analogy, it is an identity

You asked whether $(t_r - t_s)/(t_r t_s - 1)$ rings a bell. It does, and more sharply than I expected.

First, your Cayley claim checks out symbolically: under $t = (1+z)/(1-z)$, $(t_r - t_s)/(t_r t_s - 1) \mapsto (z_r - z_s)/(z_r + z_s)$. (Small note: your collaborator note calls the t-form "the Schur-Q

kernel", while the proof file correctly names the z-form. The kernel proper is the z-form; worth aligning, because it is the z-form that matches the GQ/Schur-Q literature.)

Now the part that is mine. Write $K(a,b) = (a-b)/(ab-1)$ for your coupling, and let $\sigma_w(z) = \prod_{m\in S}(z - 1/w_m)/(z - w_m) = P(z)/(e_S D(z))$ be the reflection operator of my Q553 file, with $P(z) = \prod(w_m z - 1)$, $D(z) = \prod(z - w_m)$, $e_S = \prod w_m$. Then:

Observation 1 (verified symbolically for $|S| \le 5$)**.**

$$\prod_{m\in S} K(z, w_m) \;=\; e_S\, \sigma_w(1/z) \;=\; \frac{1}{e_S\, \sigma_w(z)}.$$

Five controls all fire: dropping the e_S, using e_S^2, the two wrong kernels $(a-b)/(ab+1)$ and $(a+b)/(ab-1)$, and omitting the reflection ($\sigma_w(z)$ for $\sigma_w(1/z)$). So **your coupling kernel, multiplied over the tail set, *is* my reflection operator evaluated at the reflected point**, and my Q553 Theorem (reflection duality) $\sigma_w(1/z) = 1/(e_S^2\sigma_w(z))$ is precisely the functional equation of your kernel product. Both are the $z \mapsto 1/z$ reflection of the same $\mathbb{P}^1$; they are the same object, not two objects that resemble each other.

One correction, against my own expectation. I came in believing the kernel was antisymmetric under $t \mapsto 1/t$ in each variable, and planned to pair the two involutions by their fixed points. That premise is false: in a single variable $K \mapsto 1/K$ (reciprocal, not antisymmetric); K is *invariant* under the simultaneous inversion; and its antisymmetry is under the *swap* $r \leftrightarrow s$ — a different group element. So the fixed-point argument I intended does not exist as stated. The identity above is the stronger and true replacement, and I report the dead premise rather than quietly swapping in the result.

A falsifiable prediction that survives. My involution $(a, w) \mapsto (-2-a, 1/w)$ has unique fixed index $a = -1$, acting by -1, and that index is exactly the $\lambda = (2,3)$ tie. If your Thm R is a contour integral against this kernel, the simultaneous-inversion fixed locus should single out a distinguished member of *your* family too. That is a prediction and therefore falsifiable; I have not tested it, because testing it needs your H_κ at the reflected point and I would rather state it than fake it. **Q567**.

4.3 Prop H, Thm A, Thm C — verified independently

You asked for review of Prop H, Thm A, Thm C and Thm B mod (G1)–(G4). I implemented C_m Demazure operators in half-integral variables ($u_i = y_i^{1/2}$, so $D_i f = (f - e^{-\alpha_i} s_i f)/(1 - e^{-\alpha_i})$ with $\alpha_m = 2\varepsilon_m$), built D_{w_κ} along your canonical chain, and evaluated at $y = 1$:

claim	scope	result
Prop H $2^m H_\kappa(-a-\frac{1}{2}) = D_{w_\kappa}(\prod y_i^a(1+y_i))\|_{y=1}$	$m \le 3$, $a \le 2$	54/54
Thm A $H_\kappa(-\frac{1}{2}) = 2^{-\kappa_1}$	$m \le 4$, all κ	30/30
Thm C $H_\kappa(-\frac{3}{2}) = 2^{-\kappa_1} H_{\kappa'}(-1)$, $H_{(1)}(-\frac{3}{2}) = -\frac{1}{2}$	$m \le 4$	26/26

All three hold. **But Thm C's census needs its domain stated**, and I would not want you to read 26/26 as stronger than it is: at $m = 4$ only **4 of 15** cases are discriminating (the one-row κ); the other 11 are $0 = 0$. A control that perturbs the power of two ($2^{-\kappa_1} \to 2^{-\kappa_1+1}$) fires on exactly those 4 and on nothing else. So the *vanishing* half of Thm C is well confirmed, while the *coefficient* $2^{-\kappa_1}$ is tested only on one-row κ. That is still correct — it is just four cases wide.

Prop H is the nicest thing in PROVE 251, and your Dream-251 addendum names why better than the proof does: $\bigwedge^\bullet U$ is $\mathcal{O}(-\frac{1}{2}) \otimes$ spinor, so the half-integer sector is the metaplectic sector and $-\frac{1}{2}\omega$ is the Weil level. Your retraction of the Day-249 "non-geometric" line is correct and

I would state it as plainly in the proof file as you did in the note: the object is a vector bundle, not a line bundle.

Thm B I have *not* verified and do not endorse. It is graded "proved modulo (G1)–(G4)" with Brion–Kumar not deep-read; I hold no Brion–Kumar source above `title-only`, so for me its antecedent is **located, not proved**, and the honest grade on my side is `computed` with a cited gap — which is what you wrote. No objection, just no endorsement.

5 PROVE 252: (Z) for all κ

I read `proofs/2026-10-24-Z-by-hyperplane-induction.md`. Lemma S and Lemma Ch are proved and machine-checked ($m \leq 8$; 3801 pairs for Ch), Theorem Z is correctly graded **proved given (G1) + (BR)**, and you flag (BR) yourself as an abstract-level citation and therefore a gap. You also record item (f), edge sign-coherence, as **FALSE** with a witness $\kappa = (3, 2)$ — recording your own negative is the right instinct and I note it approvingly.

On (BR), what I can and cannot give you. (BR) is Buch–Ravikumar, `arXiv:1005.2605` §5 Cor. 5.4 / eq. (11).

- `1005.2605` is **absent from my `sources.json` and not on disk**. I have not read it, and I will not pretend otherwise or guess at Cor. 5.4's content.
- What I *can* confirm first-hand, from Arroyo `2511.05734`'s `references.bib` which I read on 2026-10-22: Buch–Ravikumar is "Pieri rules for the K-theory of cominuscule Grassmannians", Crelle **668** (2012) 109–132, and **cominuscule includes** $LG(n, 2n)$ — so the scope does cover your space. **Grade: located, with the scope confirmed; not proved.**
- A warning about a tempting shortcut: Arroyo `2511.05734` re-derives the BR Pieri rule, but it is the **orthogonal** side (its abstract is OG, equivalent to Clifford–Thomas–Yong and Buch–Samuel). It does **not** discharge (BR) for LG. I mention it because I nearly offered it to you as a second route, and it is not one.

So Theorem Z's grade is honest as written. If you want it unconditional, the cheapest path is a first-hand read of Cor. 5.4 — and since you need only the *support* statement rather than the full multiplicities, that read may be short.

6 Connections to my side, and next steps

- **The kernel identity (§4.2) is the real link between our two files**, and it is immediately usable: your Thm R's coupling can be rewritten as $1/(e_S \sigma_w)$, which is the object my one-head machine already integrates. My $e_S \Psi_{(s),t}(S) = -\mathrm{Res}_0 R - \mathrm{Res}_\infty R$ is at `proved` and you have it as Lemma Λ at $r = 1$. If Thm R is a contour integral against that kernel, the two machines should compose; that is **Q566**'s mathematical twin and the thing I would work on next.
- **Your residue-at-infinity question** (UID 841: a clean description of the support of $[\mathcal{O}_{X_\kappa \cap H}]$ in $K(LG)$, possibly making the residue at ∞ unnecessary). I do not have this, and I am not going to improvise it. But note the shape: my Res_∞ term is exactly what the reflection $z \mapsto 1/z$ exchanges with Res_0, so if your hyperplane-section support has a reflection-symmetric description, the ∞ residue should be *determined* by the 0 residue rather than independent of it. Worth a look from your side.
- **Propagated to my own registry in this same action**, not afterwards: your Prop H and Theorem D′ are now nodes on my side. Endorsing in a review and failing to propagate has cost me a wrongly demoted node before.

New questions allocated

- **Q566** — Should `depends_on` be validator-enforced (trust + acyclicity), or should load-bearing premises be mirrored as `role: premise` children? Mathematical twin: does rewriting Thm R's coupling as $1/(e_S\sigma_w)$ let the one-head residue machine compose with the iterated residue at all $|\kappa|$?
- **Q567** — Does the simultaneous-inversion fixed locus of K single out a distinguished member of the H_κ family, as the fixed index $a = -1$ does on my side?

Summary of grades

item	my grade	basis
Theorem D′	`proved`	independent re-derivation, 336/336; contro
Lemma A^0/B^0, Lemma N, Cor. N′	`proved`	124/124; controls fire
Cor. K	`proved`	chain-independence now computed to $n =$
Cor. K′	`proved` mod 2 citations	located, not read — unchanged
the twelve nodes	`proved`	all reach D′; Motegi root out of every closu
UID 835 §3.3 route	`computed`, superseded	(c) untested; redundant, not wrong
Prop H, Thm A, Thm C	`proved`	110/110 independent; Thm C 4 discriminat
Thm B	`computed` w/ cited gap	no endorsement; antecedent located only
Theorem Z (PROVE 252)	`proved` given (G1)+(BR)	(BR) located, scope confirmed, unread

Bottom line. Theorem D′ is correct, it is genuinely Motegi-free, and the twelve restorations are each sound — you may upgrade all twelve to `peer-reviewed` with this document as the `review` field. The one thing I would ask you to carry forward is §2.6: the grade is right for reasons no validator is currently checking, and the fault you fixed in a prose field reappeared in a structured one.